

Adam Zhu

(515) 708-7879 | adamzhu@andrew.cmu.edu | [linkedin.com/in/adam-zhu-cmu](https://www.linkedin.com/in/adam-zhu-cmu) | github.com/adam-zhu1 | adam-zhu1.github.io

Education

Carnegie Mellon University

Bachelor of Science in Statistics and Machine Learning; Minor in Computer Science

Pittsburgh, PA

Expected May 2029

Experience

SURF Research Intern, Research Data and Computing Office

May 2026 – Aug. 2026

National Institute of Standards and Technology (NIST)

Gaithersburg, MD

- Recovered reliable usage metrics for NIST's public data portal with a reproducible 5-stage Python pipeline (per-IP behavioral features, 4 scikit-learn classifiers, $F1 = 0.95$), revealing 51% of 190,687 requests were automated.
- Exposed a disguised scraper behind 50,000+ spoofed requests mislabeled as human by designing a second-pass behavioral detection rule, correcting ground-truth labels and improving all 4 classifiers.
- Surfaced unmet user demand (23% of 20,790 human searches returned no results) by analyzing bot-filtered search behavior, informing NIST's search and dataset-acquisition priorities.

Research Intern

Jun. 2023 – Aug. 2024

Center for Statistics and Applications in Forensic Evidence, Iowa State University

Ames, IA

- Co-authored a manuscript under review at *Forensic Sciences* evaluating camera fingerprinting for device identification by analyzing 24,000+ camera images.
- Enabled realistic handwriting-similarity analysis by developing degradation methods that produced a dataset of 100+ degraded writing samples.

Research Intern

Jan. 2024 – Jun. 2024

Virtual Reality Applications Center, Iowa State University

Ames, IA

- Showed smart audio guidance yields statistically significant cuts in evacuation time versus static and adaptive signage by fitting regression models of time-to-escape across 454 trials (227 participants) in simulated VR evacuations; co-author of a manuscript in preparation.

Projects

TrueLine — iOS Bowling Ball Tracker | Swift, SwiftUI, Core ML, Python, PyTorch, OpenCV

2025 – Present

- Developing an iOS app (App Store launch in preparation) turning a single iPhone into a ball-motion tracker: per-throw launch speed, breakpoint, and entry angle — metrics otherwise requiring a \$10k+ in-lane installation.
- Built an on-device computer vision pipeline with zero third-party dependencies by fine-tuning a YOLOv8 detector (bootstrapped from hand-labeled seed frames through a model-assisted labeling loop), exporting to Core ML with Kalman-filter tracking and Savitzky–Golay smoothing.
- Made arbitrary camera angles usable by mapping video into real-world lane coordinates via homography from a 4-corner calibration, auto-proposed by a custom gutter-line detector and refined through a magnifier-loupe drag UI.
- Verified metric accuracy (board position within 1 board, launch speed within 1–2%) by building a clip-by-clip parity harness comparing the Swift port against the Python/OpenCV prototype.

Publication

Lin, H., **Zhu, A.** and Wang, C. (2024). Statistical Tests for Proportion Difference in One-to-Two Matched Binary Diagnostic Data: Application to Environmental Testing of Salmonella in the United States. *Mathematics* 2024, 12, 741. doi.org/10.3390/math12050741

Leadership

Co-Captain

Jun. 2021 – Jun. 2025

Team Neutrino, FIRST Robotics Team #3928

Story County, IA

- Sustained the team's \$32,000 annual budget by managing a 27-sponsor corporate and community network while leading 7 sub-teams and 40+ members, contributing to multiple NASA Engineering Inspiration Awards and World Championship qualifications.

Camp Co-Designer

Summers 2024 & 2025

COR Robotics Summer Camps

Central Iowa

- Taught 100+ students in grades 3–8 foundational STEM concepts by co-designing project-based robotics curriculum (drone racing, battle drones).

Technical Skills

Languages: Python, Swift, R, Java, C, MATLAB, $\LaTeX$

Frameworks & Libraries: pandas, scikit-learn, PyTorch, OpenCV, FastAPI, SwiftUI, Core ML, AVFoundation

Tools & Data: Git/GitHub, Jupyter, matplotlib, MongoDB, REST APIs, JSON/JSON-LD, regex

Statistics & ML: Classification, Feature Engineering, Model Evaluation, Experimental Design, GLMs, ANOVA

Honors & Awards

Congressional Award (Silver Certificate), The Congressional Award Foundation, U.S. Congress

Jan. 2025

1st Place — Mathematics Category & RAYGUN Innovative Scientist Award, SSTFI

Apr. 2024

Eagle Scout Rank, Boy Scouts of America

May 2023

World Championship Engineering Inspiration Award, FIRST Robotics (1 of 6 in the world)

Apr. 2022